

BEFT: Bias-Efficient Fine-Tuning of Language Models in Low-Data Regimes

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Motivation

Fine-tuning pre-trained large language models (LLMs) for downstream tasks has gained a lot of attention over the past few years. Parameter-efficient fine-tuning (PEFT) methods have been widely studied to reduce the fine-tuning overheads [1]. Among these PEFT techniques, bias-only fine-tuning—which involves updating only the bias terms of the LLMs—provides the potential for unprecedented parameter efficiency and competitive downstream accuracy especially in low-data regimes [2]. Despite the advantages of bias-only fine-tuning, the relationship between fine-tuning different bias terms and downstream performance remains largely unclear.

Method & Findings

In this paper, we investigate the link between fine-tuning different bias terms (i.e., b_q , b_k , and b_v in the query, key, or value projections) and downstream performance. Specifically, we augment the magnitude changes with angular changes to study the link between fine-tuning different bias terms and downstream performance. **We observe that fine-tuning b_v generally leads to higher downstream performance in low-data regimes, compared to b_q and b_k .**

We also analyze the potential expressive power of the bias terms query b_q , key b_k , and value b_v for scaled dot-product attention. We find that fine-tuning b_v in low-data regimes is **sufficient**; b_k has **no effect** on improved expressiveness, whereas b_q has a **limited effect**, as shown in Fig. 1 (a).

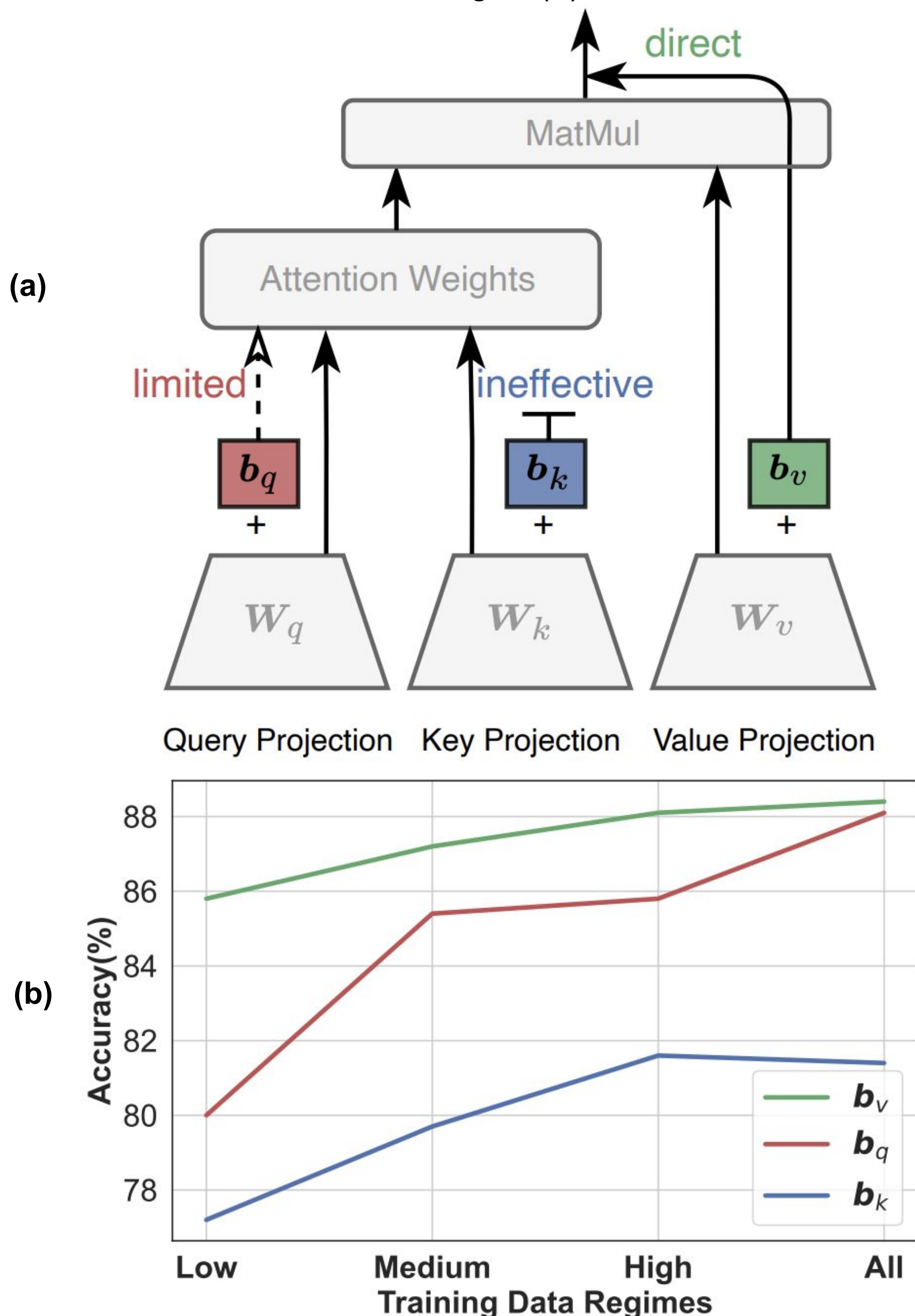


Fig. 1: Expressiveness and performance of b_q , b_k , and b_v .

Empirical Evaluation

Importance Ranking and Downstream Performance

As shown in Fig. 2 and Fig. 1 (b), we expect higher-ranked bias terms to achieve higher accuracy. Mag+Angle (ours) shows a precise and dynamic link between bias-term rankings and downstream performance across diverse data regimes.

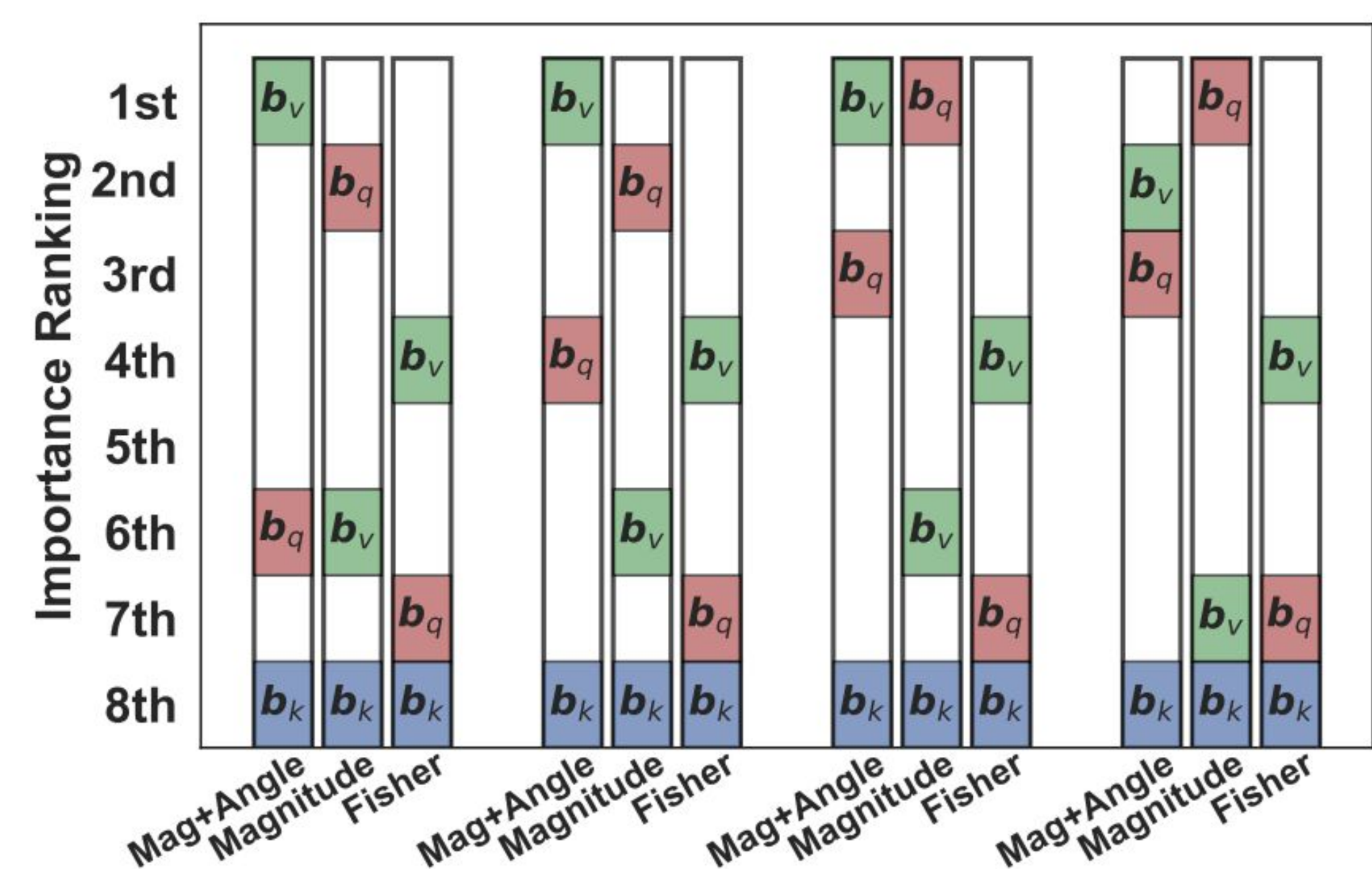


Fig. 2: Fine-tuning b_q , b_k , and b_v (BERT_{BASE} on SST-2).

BEFT can be Combined with Other PEFT Methods

Bias Term	BEFT	+LoRA	+VeRA	+DoRA
b_v	85.8%	85.0%	82.2%	86.0%
b_q	80.0%	82.6%	81.8%	83.5%
b_k	77.2%	76.9%	81.7%	76.9%

Table 1: b_v surpasses b_q and b_k in downstream performance (BERT_{BASE} on SST-2).

Extension to Bias-Free LLMs

LLMs	Adding b_v	Adding b_q	Adding b_k
LLaMA2-7B	94.9%	90.0%	68.0%
DeepSeek-Coder-Base-1.3B	76.9%	67.4%	60.3%
GPT-J-6B	92.8%	88.3%	63.8%

Table 2: Our key finding still holds (SST-2). (refer to our paper for more results)

References

- [1] Ding, Ning, et al. Parameter-efficient fine-tuning of large-scale pre-trained language models. Nature Machine Intelligence, 2023.
- [2] Zaken, Elad Ben, et al. BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models. ACL, 2022.